(1939). For an excellent recent discussion, see David Albert in *Time and Chance* (Cambridge, Mass.: Harvard University Press, 2000).

13. In fact, since the laws of physics don't distinguish between forward and backward in time, the explanation of having fully formed ice cubes a half hour earlier, at 10 p.m., would be *precisely* as absurd—entropically speaking—as predicting that by a half hour later, by 11:00 p.m., the little chunks of ice would have grown into fully formed ice cubes. To the contrary, the explanation of having liquid water at 10 p.m. that slowly forms small chunks of ice by 10:30 p.m. is *precisely* as sensible as predicting that by 11:00 p.m. the little chunks of ice will melt into liquid water, something that is familiar and totally expected. This latter explanation, from the perspective of the observation at 10:30 p.m., is perfectly temporally symmetric and, moreover, agrees with our subsequent observations.

14. The particularly careful reader might think that I've prejudiced the discussion with the phrase "early on" since that injects a temporal asymmetry. What I mean, in more precise language, is that we will need special conditions to prevail on (at least) one end of the temporal dimension. As will become clear, the special conditions amount to a low entropy boundary condition and I will call the "past" a direction in which this condition is satisfied.

15. The idea that time's arrow requires a low-entropy past has a long history, going back to Boltzmann and others; it was discussed in some detail in Hans Reichenbach, *The* Direction of Time (Mineola, N.Y.: Dover Publications, 1984), and was championed in a particularly interesting quantitative way in Roger Penrose, *The Emperor's New Mind* (New York: Oxford University Press, 1989), pp. 317ff.

16. Recall that our discussion in this chapter does not take account of quantum mechanics. As Stephen Hawking showed in the 1970s, when quantum effects are considered, black holes do allow a certain amount of radiation to seep out, but this does not affect their being the highest-entropy objects in the cosmos.

17. A natural question is how we know that there isn't some future constraint that also has an impact on entropy. The bottom line is that we